

EEMI

Recorders

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- Recorders - instruments which record electrical and non-electrical quantities as function of time.
- Electrical Quantities - V & I can be directly recorded
- Non-Electrical Quantities - Indirectly recorded, with ^(current or voltage) sensors or transducers converting to equivalent I or V

* Why Recorders needed -

It is often necessary to have permanent record or state of phenomenon being investigated.

Recording Instrument

(On the basis of type of Instrument)

- | | |
|---|--|
| <p>↓</p> <p>Analog Recorder (AC/DC Motor)</p> <p>- Input is wholly analog</p> | <p>↓</p> <p>Digital Recorder</p> <p>- if output is digital (stepper Motor)</p> |
|---|--|

↓

Graphic Recorder
(On the basis of degree of freedom)

↓

Oscillographic Recorder
↓
UV Recorder

↓

Magnetic Tape Recorder

↓

Strip chart

↓

(Synchronous Motor)
Circular Chart X-Y

↓

(On the basis of operating Principle)

↓

Galvanometric (DC Motor)

↓

Null type

↓

Potentiometric (servo Motor)

* Graphic Recorders

- chart - Displaying & storing recorded information
- Stylus - A pen & ink recorder moving in proper relation to paper.

⇒ Strip chart Recorder - records one or more variables with respect to an independent variable.

Please Refer To

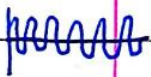
Fig - 28.48

for below given points.

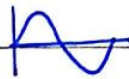
① Chart - moving vertically

② Paper Drive System - To select speed. (Driven by Synchronous Motor)

Too Slow
↓
Data - cluttered
↓
Unclear



Normal
↓
clear
Data



Too Fast
↓
Data - Spreadout
↓
Unclear



③ - Stylus - moving horizontally - in proportion to recording quantity

④ Driving System - Move stylus acc. to the recording quantity.

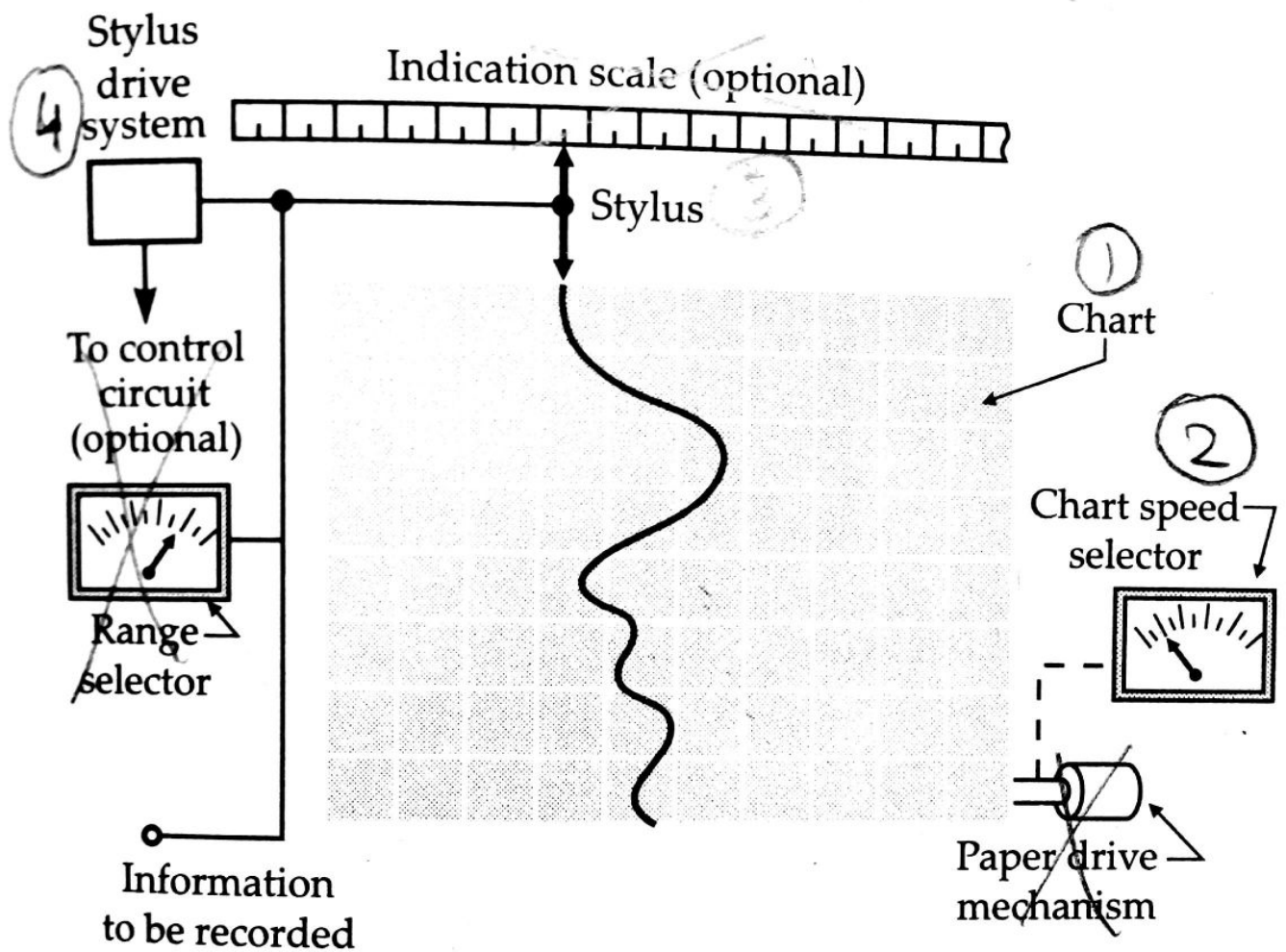


Fig. 28.48 Strip chart recorder.

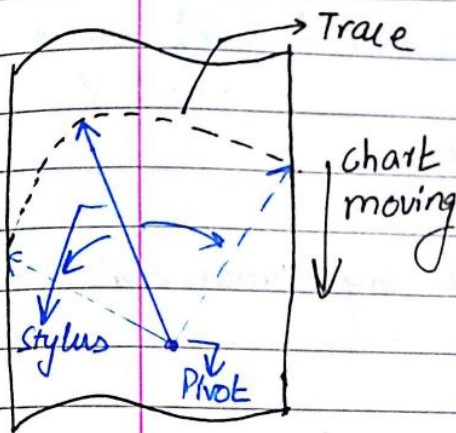
Disadvantage

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Type Paper Used	Process	Advantage	Disadvantage
① Ink Filled (Ordinary Paper used)	Stylus is filled with ink by Gravity or Capillary action	→ Wide Range of Recording Speed → Friction ↓ b/w Stylus tip & Paper	→ Ink splatter at high speed → low speed - clogs of ink at one place
② Heated Stylus (Black Paper)	Heated stylus melts thin, white wax like coating on paper	→ Reliable → High Contrast trace	→ Cannot use in process sensitive to heat
③ Chopper Bar (Pressure sensitive Paper)	V-shaped pointer pressed down by chopper bar on paper making series of marks.		→ Not continuous → Records slowly varying quantities
④ Electric Stylus	Paper with coating sensitive to current is written when current is conducted from stylus	→ ↓ Stylus Friction	→ Cost of Paper ↑
⑤ Electro-static Stylus (Electrosensitive Paper)	Stylus - produces ↑ Voltage discharge on paper	→ ↑ Speed Response	→ ↑ Speed resp
⑥ Optical Marking Method (Photosensitive Paper)	Stylus throws beam of light on paper	→ ↑ change speed	→ Paper cost ↑

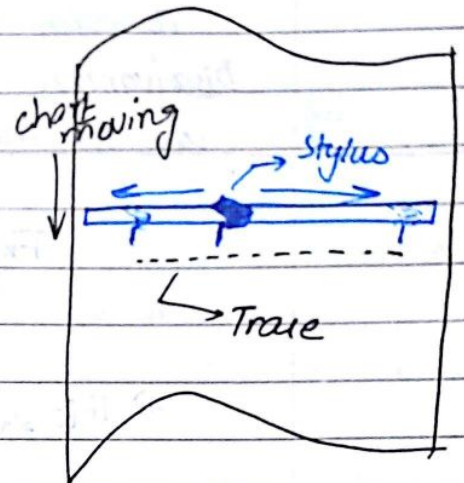
Tracing System

Curvilinear



- ⇒ Stylus is mounted on a pivot which moves in a arc
- ⇒ This system may be used on PMMC galvanometers.
- ⇒ Used in tracing helical path followed by $e^\ominus$ in magnetic field.

Rectilinear



- ⇒ Time varying quantity is $\perp$ to the axis of time.
- ⇒ system ^(stylus) may be actuated by drive cord, self balancing potentiometer etc.

* Galvanometric Type Recorder-

This strip chart recorder operates on deflection principle.

measuring quantity applied $\Rightarrow$ Proportioned current generated

Stylus $\leftarrow$ Torque $\leftarrow$ Current passed through coil
Actuated applied

- ⇒ To control the motion of stylus, $\Rightarrow$ controlling torque exerted by spring.

- ⇒ To trace the quantity $\Rightarrow$ appreciable torque required.
large moving $\downarrow$ coil situated in a strong magnetic field.

⇒ Also, instrument should be critically damped ⇒ so no significant

But this produces ← overshoot
↓ slow response

Recorder not useful for

recording fast variation in current/v/power ⇒ Average value Recorded

Disadvantage

⇒ Low I/P Impedance & Limited sensitivity

Fig - 28.49 ⇒ Shows PMMC, i.e. galvanometric recording device

⇒ This shows a curvilinear system of tracing

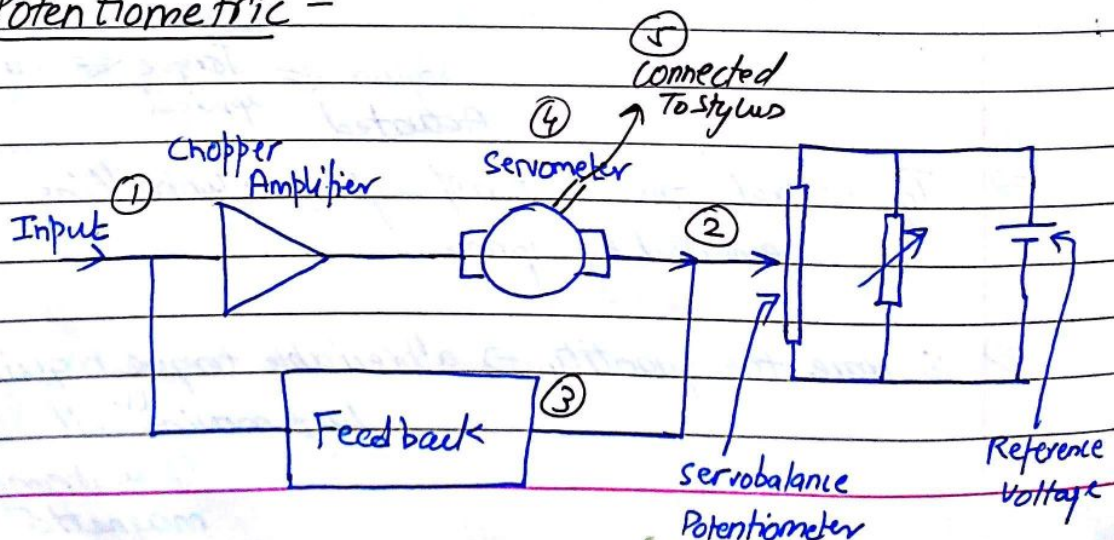
Fig - 28.50 ⇒

Shows a galvanometer type recorder which has a restricted span of movement that in essence we can say it shows a rectilinear system of tracing.

* Null Type Recorder

- works on the principle of comparison.

⇒ Potentiometric -



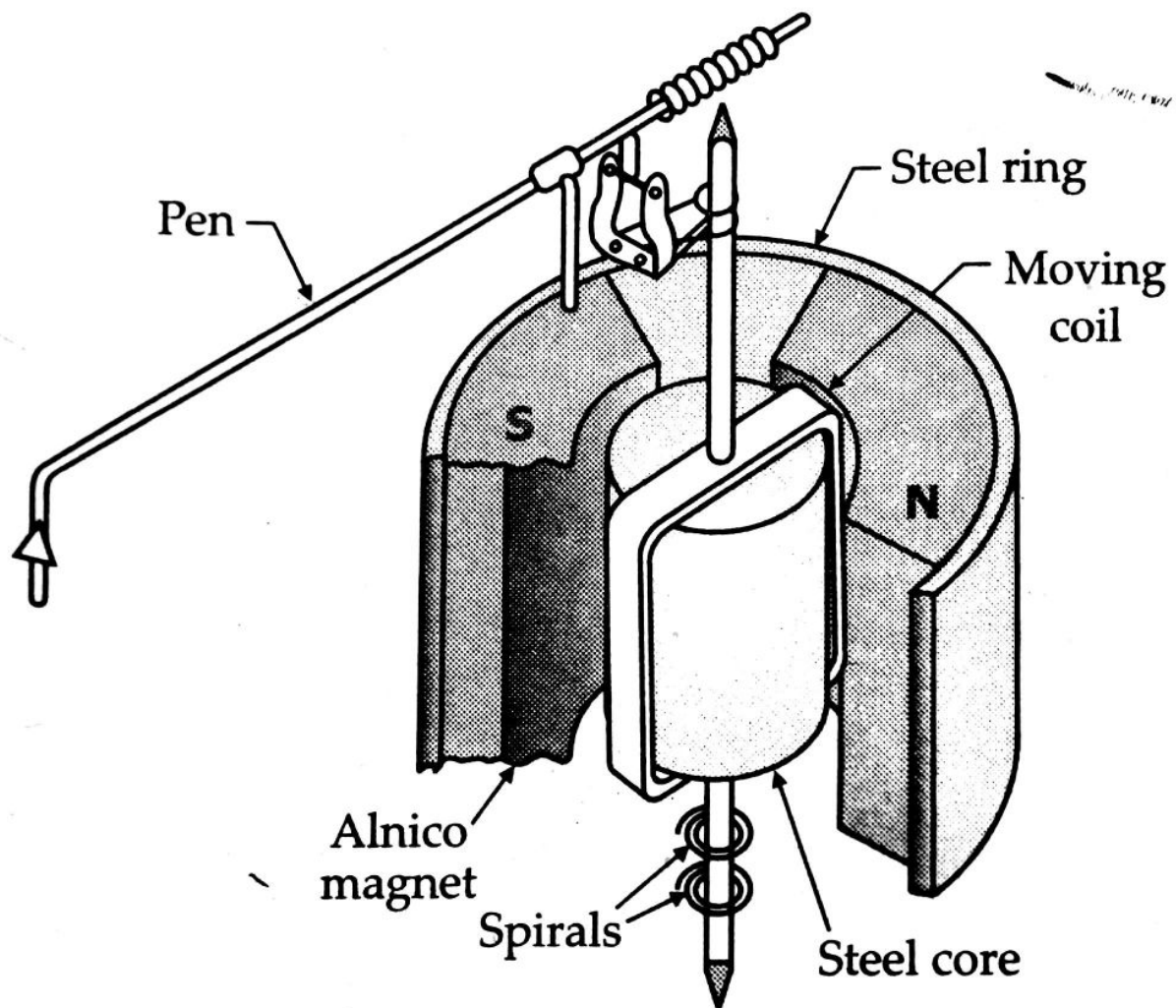


Fig. 28.49 Cut-away view of moving coil element.

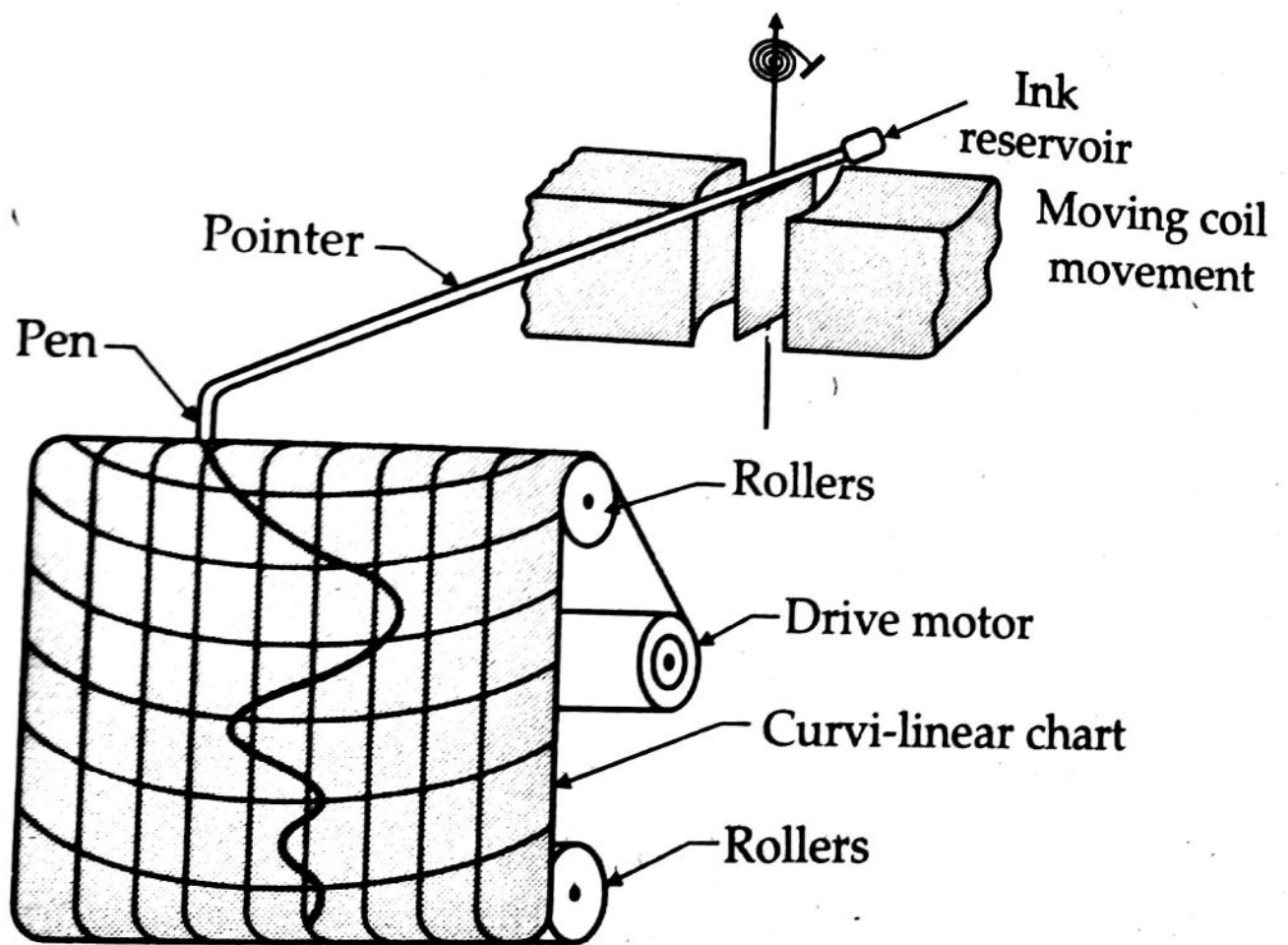


Fig. 28.50 Components of a galvanometer type recorder.

- ⇒ The self balancing potentiometer type of instrument consists of bridge circuit.
- ⇒ Initially bridge is adjusted ⇒ servomotor & amplifier are in balance
stylus is at reference line (mostly 0)

* Working - (Refer to fig. for points)

- ① Signal is fed to amplifier
- ② Output causes the servomotor to drive to a balancing potentiometer
- ③ which in turn refers a feedback voltage to amplifier i/p.
- ④ When two signals are equal & opposite, system balances and servomotor stops.
- ⑤ A pen is attached to motor (potentiometer mechanical drive)
pen will show the proportional value of i/p signal.

* This arrangement remove disadvantage of galvanometer

As at i/p amplifier is connected which has $\uparrow$ i/p impedance ⇒ helps in removing loading effect

as it has $\uparrow$ i/p impedance (which is ∞)
+ high sensitivity. (at balance point)

ex - used for recording & control of process temperature.

* Circular Chart Recorder -

- ⇒ Records data in circular format.
- ⇒ Paper spun in circular motion
- ⇒ Pen - deflected in proportion to varying signal resulting in circular chart.
- ⇒ Normally chart rotation $\Rightarrow$ slow $\Rightarrow$ 1 rotation in 1 day
- ⇒ Ex - ^{Recording} Variation of temp. $\Rightarrow$ over months $\Rightarrow$ (Not on rectilinear to avoid wastage)

Fig 15.5 $\Rightarrow$ shows Industrial Circular Recorder with time axis or time arcs in concentric circles.

Analog Recorders

(On the basis of no. of parameters recorded)

Single Point Recorders

- Record only 1 changing variable with respect to time.
- Usually single continuous Curve.

Multi-point Recorders.

- Records several changing I/P at same time.
- Ex - In Process industry
Necessary to simultaneously record variables like temp, pressure, flow rate etc.
- May have monⁿ of 24 I/P with 6 colours.

* X-Y Recorders - Is an instrument which gives a graphic record of the relationship b/w 2 variables.

- ⇒ In strip chart recorder, self balancing potentiometers are used. These self-balancing potentiometers plot emf as function of time.

In XY-Recorders, emf is plotted as function of another emf.

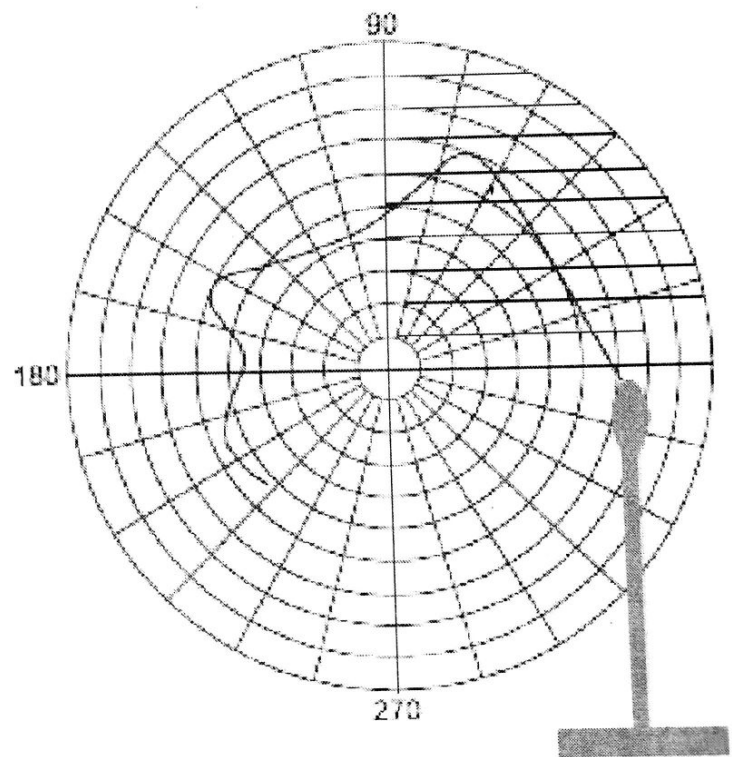


Figure 15.5 *Circular chart recorder*

Recording



Paper ~~stationary~~ stationary

⇒ 1 self-balancing potentiometer circuit moves recording pen ⇒ X direction

⇒ Other self-balancing potentiometer circuit moves same recording pen ⇒ Y-direction.



Paper ^{MOE} stationary

⇒ One self balancing potentiometer control position of paper

⇒ other ⇒ position of pen.

⇒ Many times ⇒ Emf may be of transducer measuring
displacement, force, pressure, strain, etc.

One physical quantity may be plotted against other physical quantity.

Please refer to fig - 28.53 (X-Y Recorder)
for below given points

- ① Signal enters each of 2 channels
- ② Signals are ^{attenuated} amplified in proportion to full scale deflection to maximum allowed input.
- ③ Passes through a balance circuit which compares to reference voltage (ex-like potentiometer circuit).
- ④ This error signal (i.e. the difference b/w i/p & reference voltage) fed to chopper ⇒ convert dc to AC signal.
- ⑤ Signal Amplified ⇒ to actuate servomotor
- ⑥ servo-motor attached to pen to move in proportion to signal.

This whole process takes place in both axes simultaneously. ⇒ get record of one variable w.r.t another.

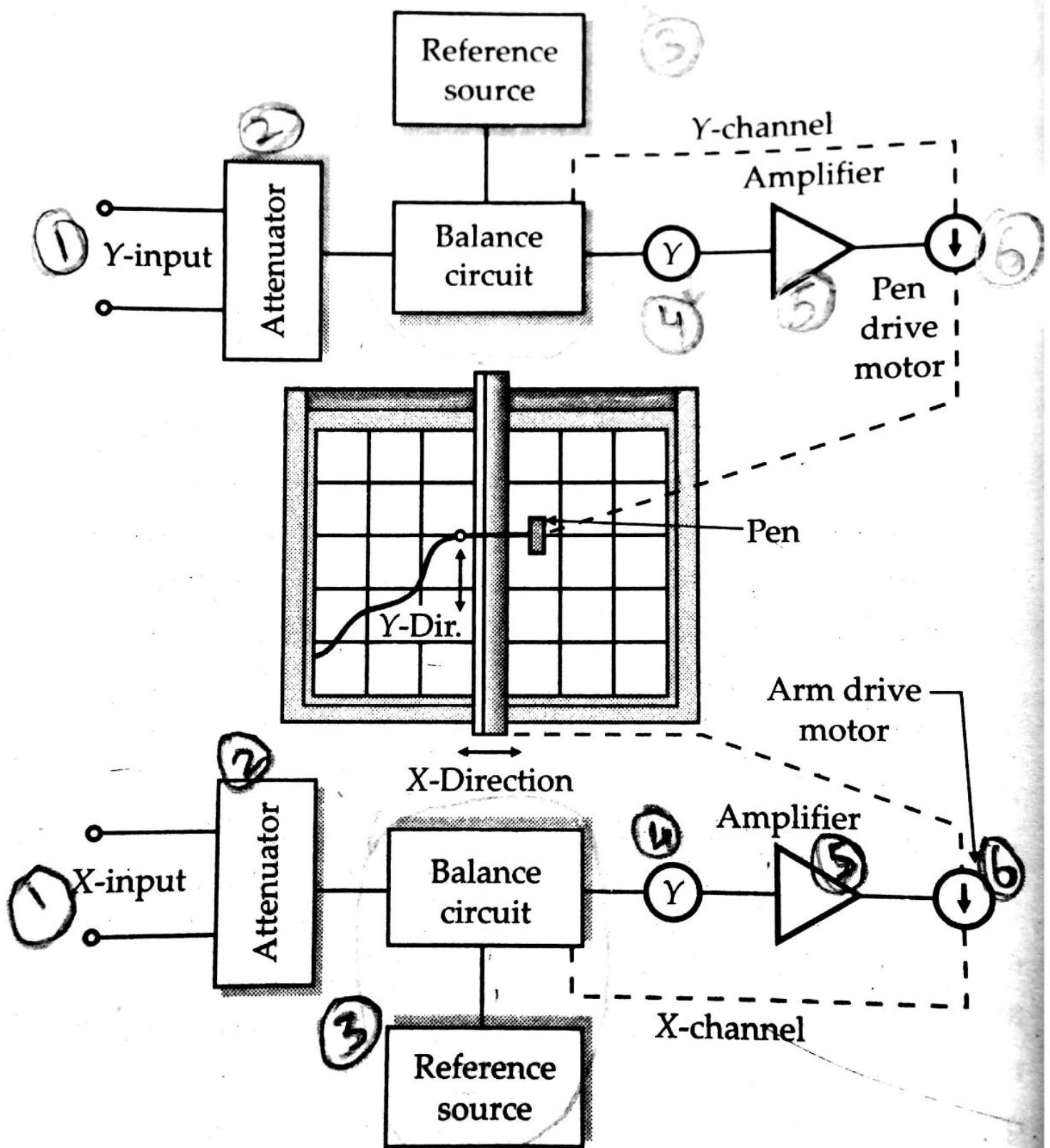


Fig. 28.53 X-Y Recorder.

* Ex - Used for

- speed torque characteristics of motor
- Plotting characteristics of zener diodes, rectifiers, transistors.
- Plotting hysteresis curve.

* Ultraviolet Recorders -

These are electro-mechanical oscillographic recorder which uses a source of ultraviolet light and ultraviolet sensitive paper used for plotting a trace.

(Current proportional to input signal) Refer to Fig - 28.54 for working

- ① Current is passed through moving galvanometer coils.
- ② The coils get deflected under the influence of magnetic field of permanent magnet.
- ③, ④, ⑤ UV light falls on mirror is deflected and projected on U.V. light sensitive paper through lens and mirror system.
- ⑥ Paper driven past the moving light spot and thus a trace of variation of current with respect to time is produced.

regulation curves of power supplies.

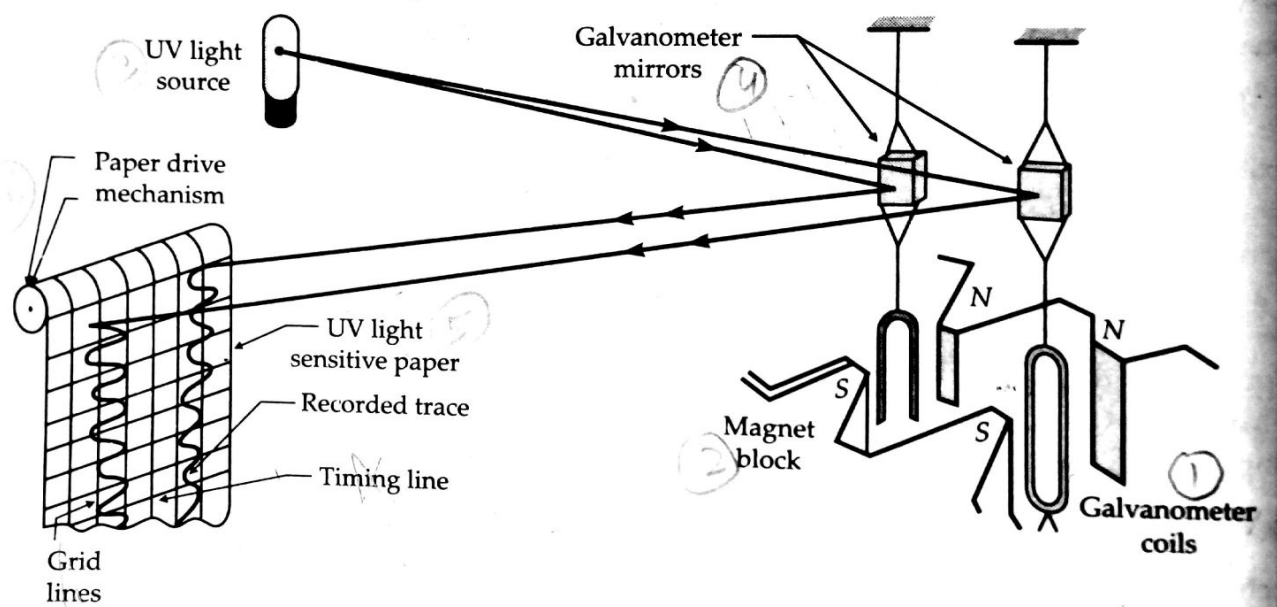


Fig. 28.54 Ultraviolet (UV) recorder.

* Magnetic Tape Recorders-

- ⇒ Many time necessary to record data in such a way that they can be retrieved or reproduced in electric form again.
- ⇒ Till now all $\downarrow$ frequency recorders but magnetic tape recorders have response characteristics which enable them to be used at $\uparrow$ $\downarrow$.

Refer fig - 28.55 (Next Page)

For below points.

① Recording head -

This device responds to an electric signal in such a manner that a magnetic pattern is created in magnetic ~~medium~~ medium. A fine air gap of length 5-15 μm is shunted by passing magnetic tape.

A current in coil causes flux of same shape to bridge the air gap & hence to pass through the magnetic tape; thereby magnetizing the iron oxide particles as they pass the gap.

The state of magnetization of oxide as it leaves the gap is retained (due to hysteresis).

Thus actual recording occurs at trailing edge of the gap.

- ⇒ Signal recorded on the tape appears as a magnetic pattern dispersed in space along the tape, similar to original coil current variation with time.

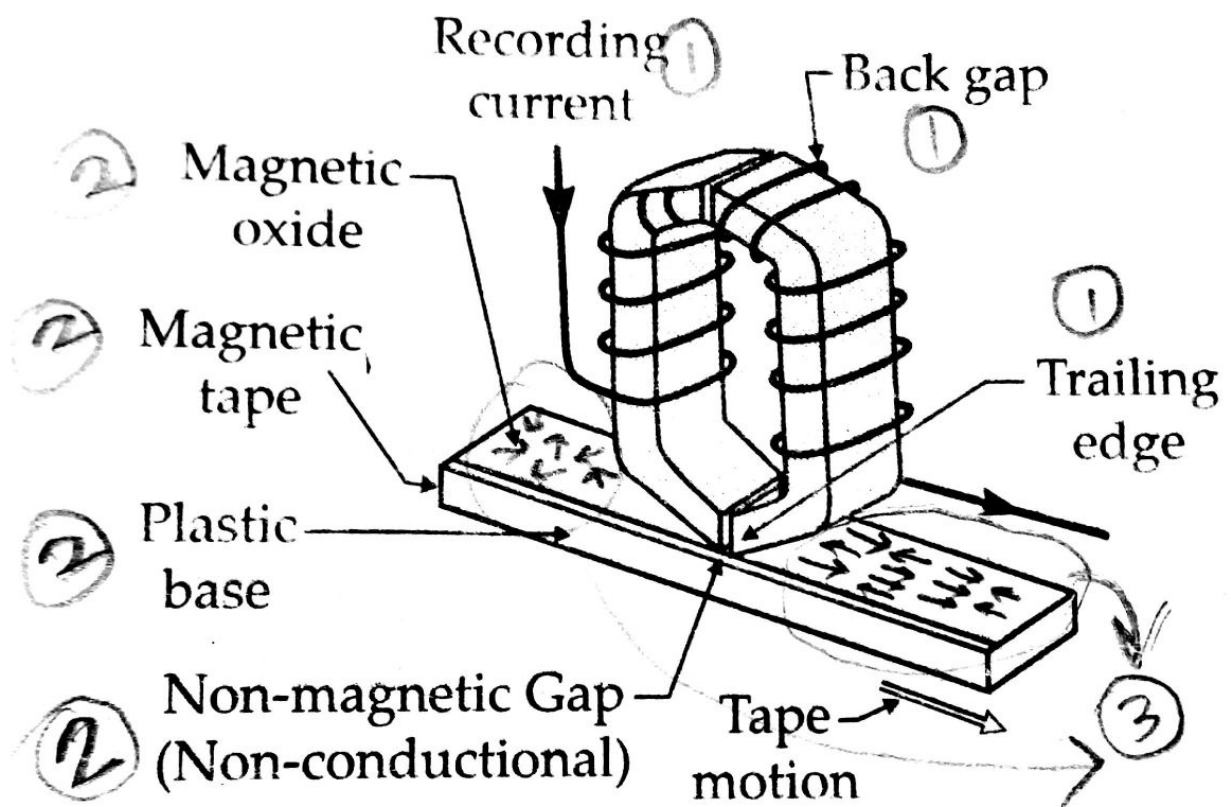


Fig. 28.55 Magnetic tape recording head.

③ Shows the alignment of domains in specific order after being processed from by the magnetic head.

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② Magnetic tape - composed of fine magnetic iron oxide particles on plastic ribbon.

⇒ A reproducing head similar in appearance to recording head is used to detect the magnetic pattern stored in them & convert it back to original electrical signal.

⇒ Advantages of Magnetic Tape Recorder -

→ Have wide range of recording (dc to several MHz)

→ Magnitude of electrical I/p signal ⇒ stored as magnetic memory

↙ reproduced whenever desired.

→ Record signal is immediately available
No time lost in processing.

→ Signal can be played back / reproduced any no. of times without loss of signal.

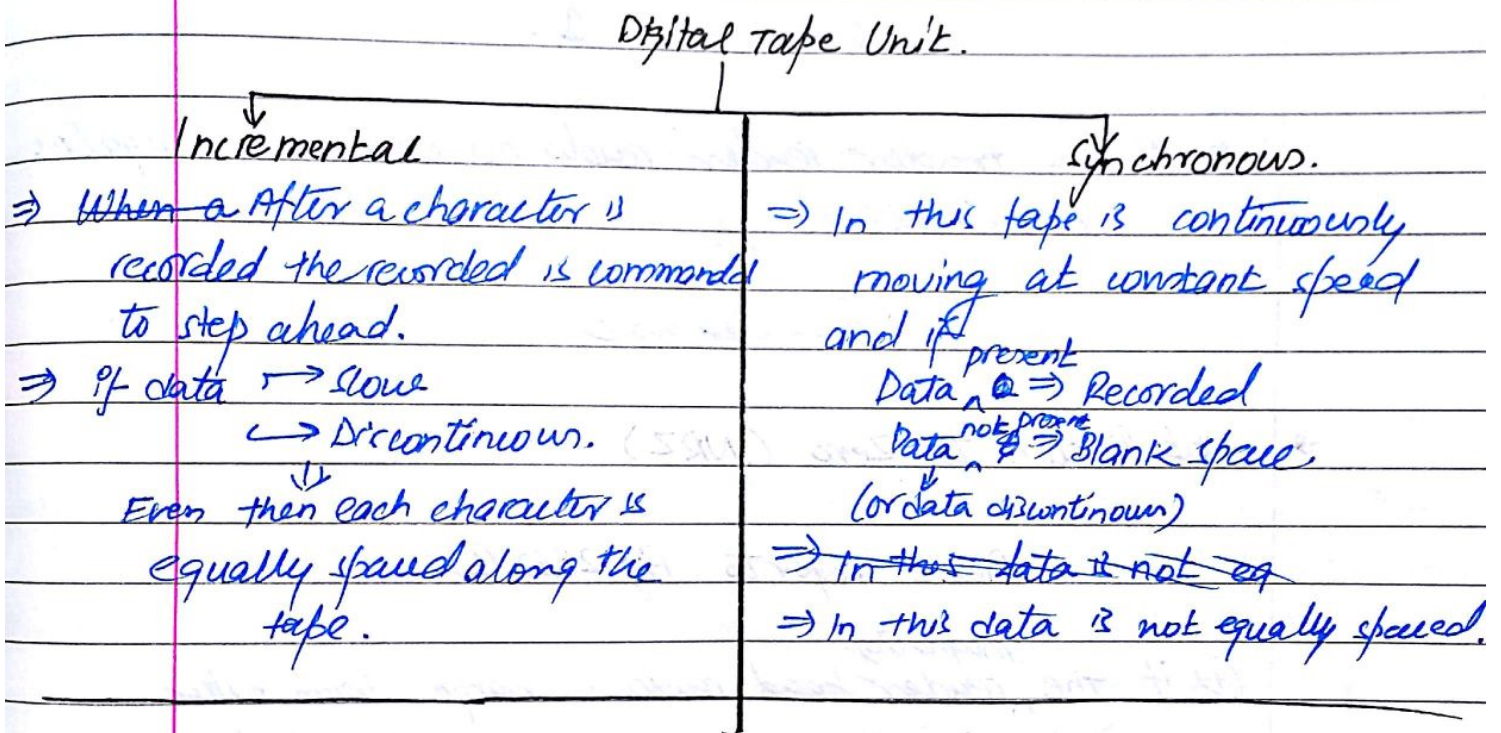
→ Tape can be easily erased & reused.

/Playing
⇒ Reproducing Action -

Some tape when passed through a reproduce/playback head produces variation in the reluctance of winding thereby inducing a voltage in winding dependant upon the direction of magnetization and its magnitude on magnetic tape.

* Digital Tape Recorders-

Digital tape recorder are generally used as storage devices in data processing applications.



⇒ Digitally store - characters are represented on magnetic tape by combination of 1-bits in binary code ⇒ 0 and 1.

Storing Process.

Return to Zero
(RZ)

Not Return to Zero
(NRZ)

* Return to Zero (RZ)

Refer to Fig - 28.62 (a)

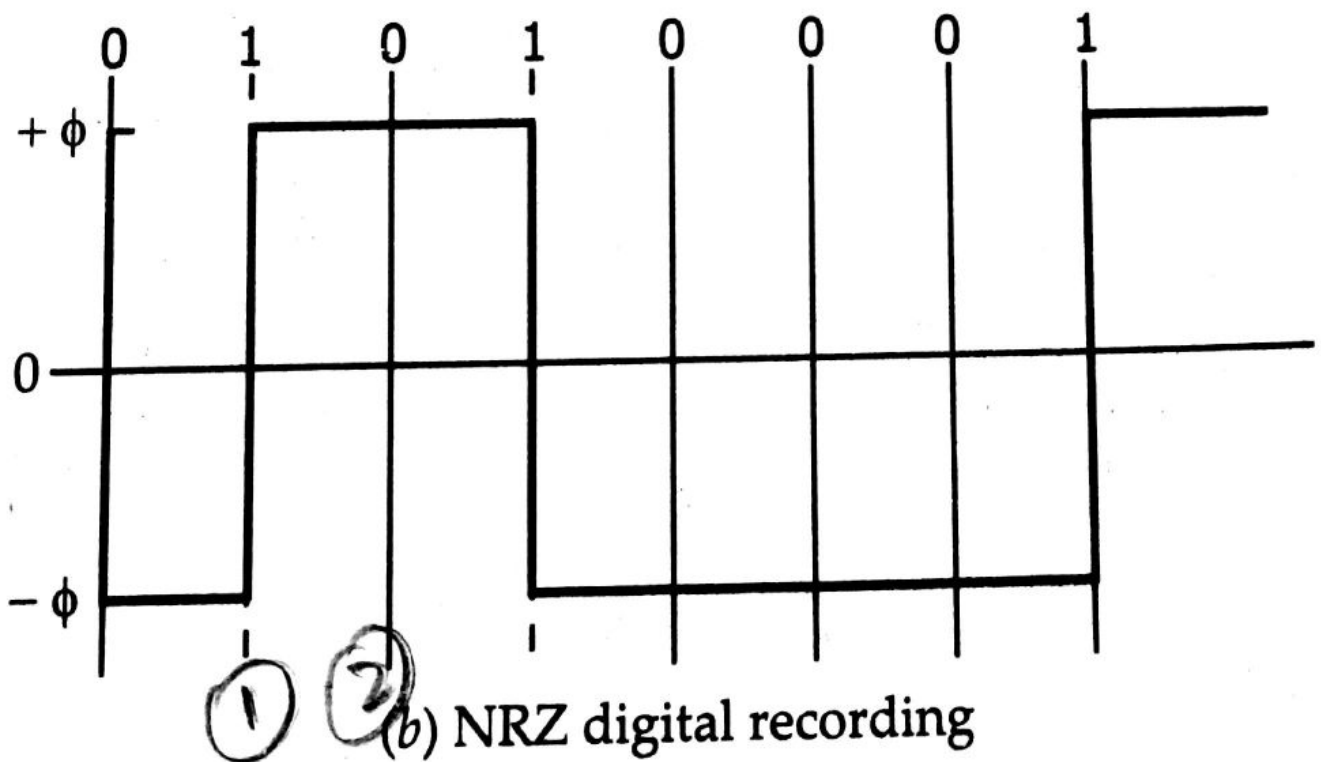
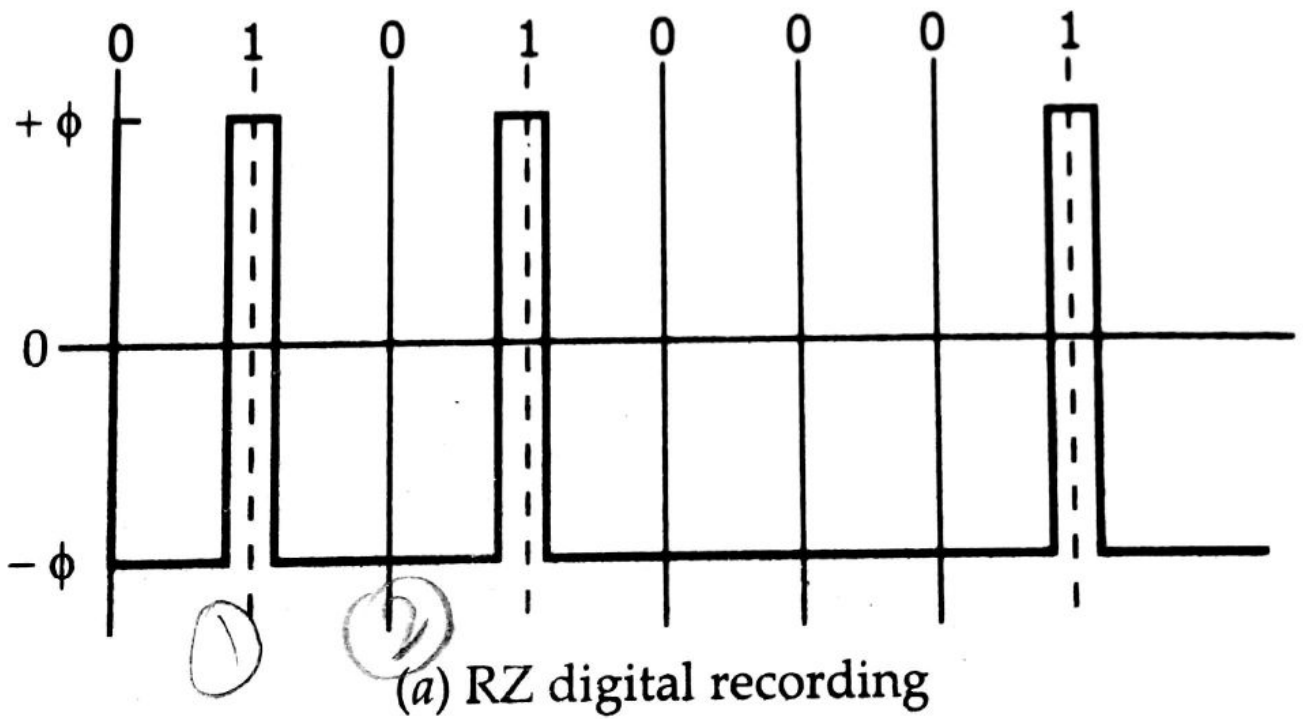


Fig. 28.62 Digital tape recording.

① If the reader reads a positive magnetized space at place with ~~negative~~ magnetization on the tape before or after this place on the magnetic tape



Recorded as 1.

② if the ~~header~~ reader reads no change in negative magnetization ⇓

Recorded as 0

* NRZ Return to Zero (NRZ)

Please refer to Fig 28.62 (b)

① if the ^(magnetic) reader head reads a change from either
 $+ve \rightarrow -ve$ or $-ve \rightarrow +ve$ magnetisation $\Rightarrow$ Recorded 1

② if no change in state $\Rightarrow$ Recorded 0